

Uncertainty estimation in Deep Learning models

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Outline

Quantifying uncertainty in model predictions

Predictions of machine learning models, like any other models, may be uncertain. Quantifying this uncertainty for medical imaging applications can be crucial. How do we do this for deep learning models?

- 1 The problem
- 2 Mathematical treatment
 - Posterior distribution
 - Approximations
 - Evaluation
- 3 Examples
- 4 Unintuitive behavior in high dimensions

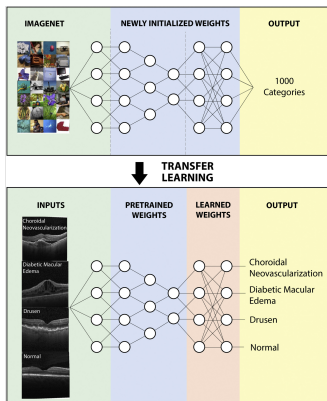
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In classification



[Kermany et al., 2012]

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Tomography images of the eye.

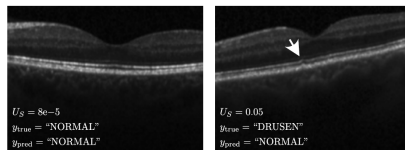


Fig. 1: Uncertainty estimation in retinal OCT scan. Left: Correctly predicted scan results in low prediction uncertainty U . Right: False prediction correlates with high prediction uncertainty. In this scan, the disease characteristics are only weakly present (white arrow), which makes the diagnosis challenging [9].

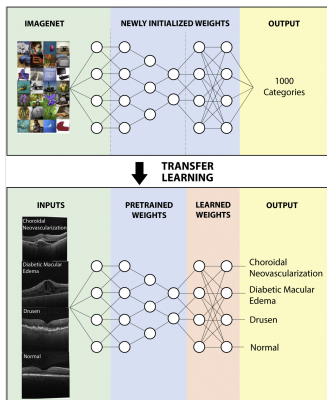
[Laves et al., 2019]

Some cases are very difficult to identify, even for experts.

The disease effect can be very subtle.

Image may not provide enough information for a clear classification.

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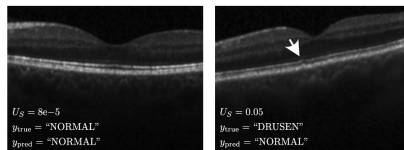


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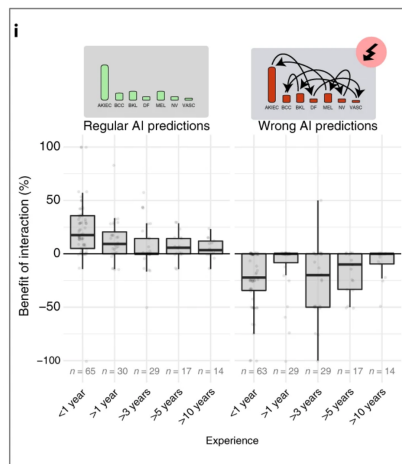
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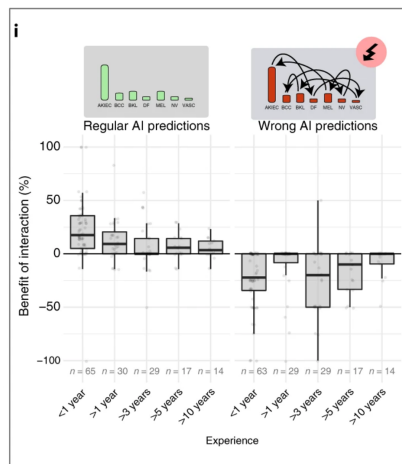
It is important to acknowledge these challenges and assign high uncertainty.

Why is it important?



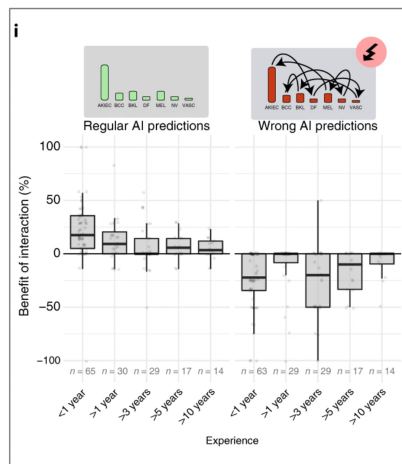
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- Deep learning methods benefit raters for classifying skin disease.
- Classes predicted with high confidence are taken more into account by raters.

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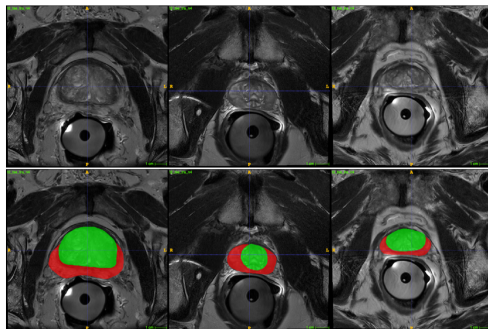
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- In the plot they show that if the model provides a very high confidence to an inaccurate class, raters changed their opinions in favor of model's prediction.
- It is crucial not to be highly confident when you are wrong.

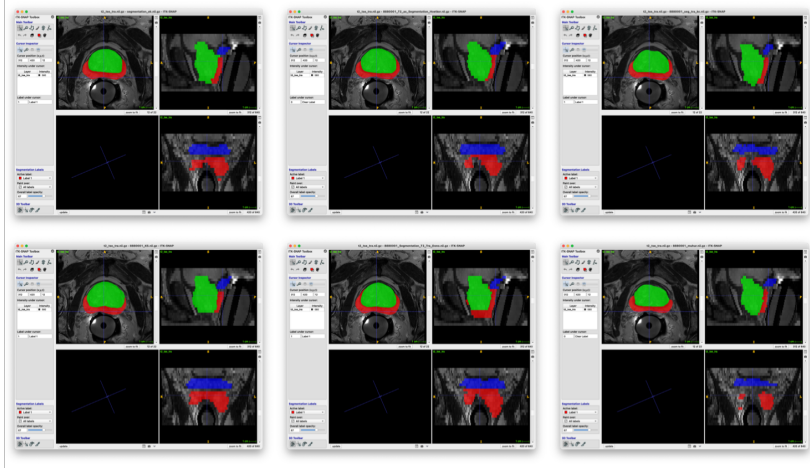
In segmentation



While the images show clearly where the objects are, boundaries are much less clearly defined due to limitations in the image resolution and contrast.

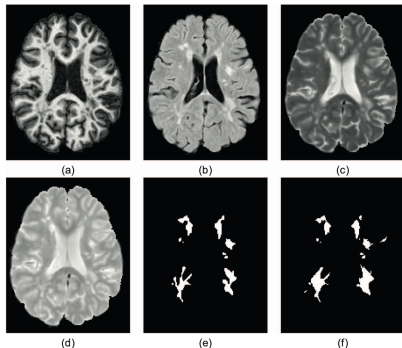
Effectively, one uses their prior knowledge in delineating the boundaries when they are segmenting the images.

Prior information can be slightly different for different people



Raters cannot agree on the boundaries [Becker et al. 2019].
The same image is segmented manually by 6 raters.

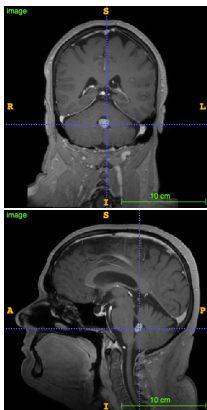
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Images show four input sequences and segmentations of two expert raters.
[Carass et al. 2017]

- Imaging is crucial in monitoring a number of diseases.
- Uncertainty in segmentation leads to uncertainty in quantification of the lesion (disease) load on the anatomy.
- In the longitudinal setting, differences in global and local lesion load is crucial for assessing treatment efficacy.
- For correct assessment, uncertainty must be taken into account.

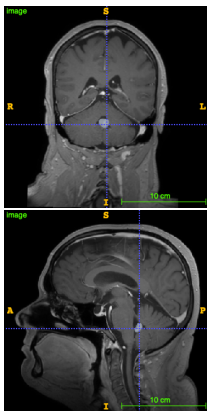
In detection



Images show coronal and sagittal views of brain MRI with possible metastases.

- Imaging is primarily used for diagnosis.
- Detecting abnormalities is a crucial step.
- Identifying clearly normal and clearly abnormal anatomy can be tremendously helpful for improving workflow efficiencies in clinical practice.

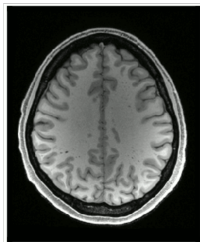
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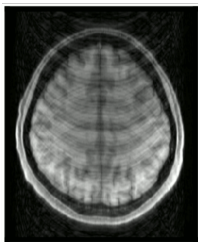
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- Imaging is primarily used for diagnosis.
- Detecting abnormalities is a crucial step.
- Identifying clearly normal and clearly abnormal anatomy can be tremendously helpful for improving workflow efficiencies in clinical practice.
- Some abnormalities are difficult to discern from normal anatomy.
- They require human readers, and identifying them by quantifying their high uncertainty is important.

In reconstruction from undersampled data



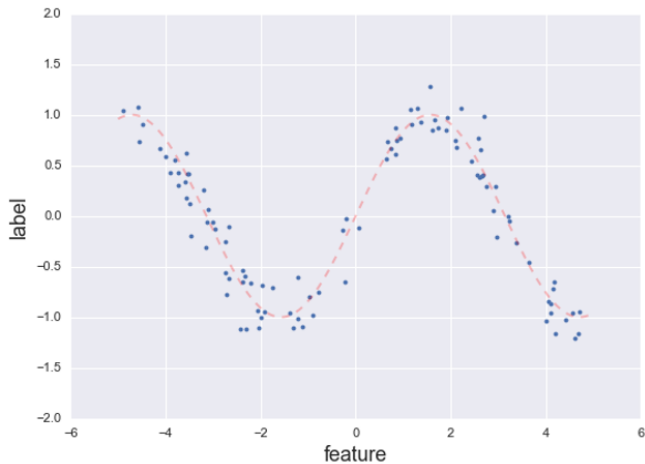
Fully sampled



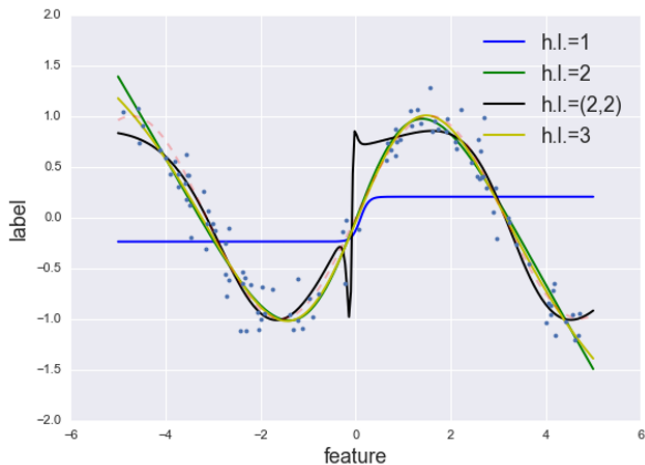
Undersampled by
a factor of 5

- Image reconstruction from undersampled data is an ill-posed problem.
- Missing image information needs to be complemented using prior information.
- Prior information can fill the image in multiple ways.
- Uncertainty in reconstruction leads to uncertainty in morphological measurements.
- Models should not provide confident predictions for information missing from data.

Generally in model fitting

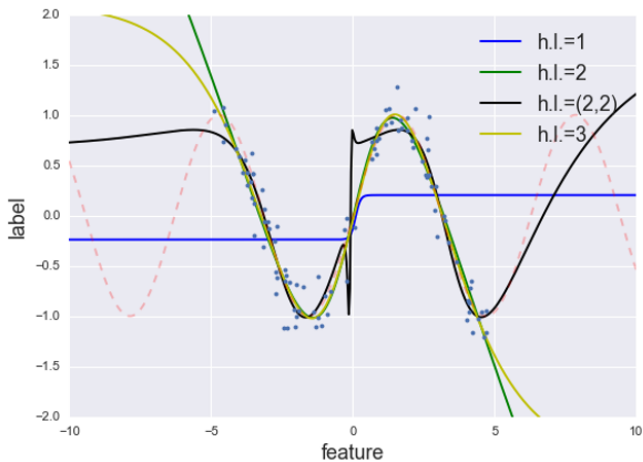


Generally in model fitting



Uncertainty in predictions due to model's architecture and parameters in interpolation.

Generally in model fitting



Uncertainty in extrapolation is even further amplified due to architecture and parameters

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Notation

We use the usual notation that we got accustomed so far and we will add one more notation

x	features
y	labels
$f(x; \theta)$	a neural network model
θ	parameters of the network
$\mathcal{D}$	a training data set

Prediction happens with

$$y \approx f(x; \theta^*)$$

and training happens with

$$\theta^* = \arg_{\theta} \min \mathcal{L}(\mathcal{D}; \theta)$$

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There are many good resources written by David Mackay, Christopher Bishop, Kevin Murphy, ...

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I have to say the last item is not necessarily a source of uncertainty but of error that may be different for each sample.

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This is important

This topic is very important. You should see it and think about it despite the challenges and disagreement.

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All of these integrations are challenging.

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- $\int_{\theta}$ requires going through all possible θ and evaluating the model.

This actually sounds feasible.

But it is also challenging because it is not trivial to get $p(\theta|\mathcal{D}, \mathcal{M})$ since θ is often very high dimensional.

What happens when we train and predict?

Before we see approximations, we should interpret from the perspective of the probabilistic model what we do when we train networks and then do inference “in the usual way”. Let us assume that we do not do any hyper-parameter optimization. Similar story holds also with hyper-parameter optimization.

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Training:

$$\begin{aligned}\theta^* &= \arg_{\theta} \min \mathcal{L}(\mathcal{D}; \theta) \\ p(\theta | \mathcal{D}, \mathcal{M}) &\approx \delta(\theta - \theta^*)\end{aligned}$$

where $\delta()$ is the Dirac distribution, i.e., $\int \delta(z) dz = 1$ and $\int \delta(z - z_0) g(z) dz = g(z_0)$.

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Inference:

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We first fix one $\mathcal{M}$ and one $\mathcal{D}$.

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Expected prediction or any other sample

$$f(x; \theta^*) = \mathbb{E}_{y|x, \mathcal{M}, \mathcal{D}}[y]$$

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$$p(y|x) = \int_{\mathcal{M}} \int_{\mathcal{D}} \int_{\theta} p(y|x, \theta, \mathcal{M}) dp(\theta|\mathcal{D}, \mathcal{M}) dp(\mathcal{D}) dp(\mathcal{M})$$

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- 1 Sampling instead of integration
- 2 Approximating posterior distributions

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Instead of integrating all the variables, we can do **ancestral sampling** using model.

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This looks good, but in reality even sampling from these distributions are challenging due to various reasons. So we need to combine this with approximations of posterior distributions and also priors.

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Sampling option is heavily used for ensembling methods that show great prediction quality and estimate uncertainty.

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Approximate sampling of different models is often done

- by selecting a number of preferred models, e.g., a fully convolutional network and a U-Net for segmentation.
- by generating new models around a preferred model, e.g., adding layers to channels of a U-Net, changing architecture in the encoding path, removing skip connections.

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Not feasible to cover all possible models. Cost of sampling multiple models is the increased time and energy needed to train all of them.

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Use Bayes' theorem to better understand this and link it to the cost functions.
We will assume $\mathcal{M}$ is given.

$$p(\theta|\mathcal{D}, \mathcal{M}) = \frac{p(\mathcal{D}|\theta, \mathcal{M})p(\theta|\mathcal{M})}{p(\mathcal{D})},$$

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There are two parts of the numerator:

- 1 $p(\theta|\mathcal{M})$ is the prior distribution for the model parameters.
- 2 $p(\mathcal{D}|\theta, \mathcal{M})$ is the likelihood, i.e., how well the parameters and the model satisfy the data.

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The usual training is the maximum-likelihood estimation of the parameters

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If there is a regularization over the weight parameters, e.g., weight decay, then the usual training can be seen as the **maximum-a-posterior (MAP) estimation** of the parameters.

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Note that MAP estimation is the same as the maximum-likelihood estimation if there is no regularization term, i.e., $p(\theta|\mathcal{M}) = \text{const.}$

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 $\Rightarrow$ approximate factorization of the sort $p(\theta|\mathcal{D}, \mathcal{M}) \approx \prod p(\theta_i|\mathcal{D}, \mathcal{M})$ may not be very good approximations.

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2. Approximate sampling with by solving $\theta^s = \arg_{\theta} \max \log p(\mathcal{D}|\theta, \mathcal{M}) + \log p(\theta|\mathcal{M})$ with a different initialization for every sample, e.g., $\theta_0^s \sim p(\theta_0|\mathcal{M})$ where a simple factorized distribution can be used as a prior.

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You can imagine combining one of the first two with the third.

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where σ needs to be fixed otherwise you would have trivial solutions. Or with priors over θ , e.g.,

$$\arg_{\mu, \sigma} \max \log p(\mathcal{D}|\theta^l, \mathcal{M}) - D_{KL}[\prod_i \mathcal{N}(\theta_i; \mu, \sigma) || \prod_i \mathcal{N}(\theta_i; \mu_0, \sigma_0)],$$

where $\theta^l \sim \mathcal{N}(\theta_i; \mu, \sigma)$.

Training with approximation for option 3

Instead of using an approximation fixed after MAP estimation

$$p(\theta|\mathcal{D}, \mathcal{M}) \approx \prod_i \mathcal{N}(\theta_i; \theta_i^*, \sigma),$$

You can train the model with the distributional approximation without any priors over θ , e.g.,

$$\arg_{\mu} \max \log p(\mathcal{D}|\theta^l, \mathcal{M}), \quad \theta^l \sim \prod_i \mathcal{N}(\theta_i; \mu, \sigma)$$

where σ needs to be fixed otherwise you would have trivial solutions. Or with priors over θ , e.g.,

$$\arg_{\mu, \sigma} \max \log p(\mathcal{D}|\theta^l, \mathcal{M}) - D_{KL}[\prod_i \mathcal{N}(\theta_i; \mu, \sigma) || \prod_i \mathcal{N}(\theta_i; \mu_0, \sigma_0)],$$

where $\theta^l \sim \mathcal{N}(\theta_i; \mu, \sigma)$.

Note that D_{KL} is the Kullback-Leibler divergence and the final argument in the optimization corresponds to the Evidence Lower Bound (or variational lower bound) for $p(\mathcal{D}|\mathcal{M})$.

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Also note that in the model

$$p(y, \theta, \mathcal{D}, \mathcal{M}|x) = p(\mathcal{M})p(\mathcal{D})p(\theta|\mathcal{D}, \mathcal{M})p(y|x, \theta, \mathcal{M})$$

this is the only term modeling the dependency of y on x . Therefore, it is the best place to integrate the fact that multiple y 's can correspond to a single x .

Beyond the simple model for $p(y|x, \theta, \mathcal{M})$ - I

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$$p(y|x, \theta, \mathcal{M}) \triangleq \mathcal{N}(y; f(x; \theta), g(x; \theta)),$$

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For the simple Gaussian case, this would mean an L_2 norm and a term for the standard deviation

$$\max_{\theta} \left\{ -\log(g(x; \theta)) - \frac{1}{2} \frac{(y - f(x; \theta))^2}{g(x; \theta)^2} \right\}$$

Beyond the simple model for $p(y|x, \theta, \mathcal{M})$ - I

For a model that predicts images, i.e., a prediction at each pixel, this can be modified

$$p(y|x, \theta, \mathcal{M}) \triangleq \prod_i \mathcal{N}(y_i; f_i(x; \theta), g_i(x; \theta)),$$

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where a factorized distribution is assumed at the end.

If statistical dependence between different output pixels need to be modeled, more complicated models can be used, e.g., compound distributions with lower dimensional latent space

$$p(y|x, \theta, \mathcal{M}) \triangleq \int_z p(y|z, x) p(z|x) dz,$$

where similar $p(z|x)$ and $p(y|x, z)$ are modeled by networks (as in Variational Autoencoders) [Kohl et al., 2018].

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The second modeling does not predict a probability distribution but generates different samples given the input [Wang et al. 2018]. Effectively, it does not explicitly model a probability distribution.

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$$y^l = f(x, z^l; \theta), \quad z^l \sim p(z),$$

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where z is a random input.

To make sure that different samples of y^l make sense, it is common to use a distributional loss in the training, e.g.,

$$\min_{\theta} \max_{\psi} \sum_n L(y_n, f(x_n; \theta)) \\ + \lambda \left(\mathbb{E}_{y \sim p(y)} [\log D(y; \psi)] + \mathbb{E}_{y=f(x, z^l; \theta), x \sim p(x), z^l \sim p(z)} [\log(1 - D(y; \psi))] \right)$$

Evaluation

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$$D_{GED}^2 = 2\mathbb{E}[d(s, y)] - \mathbb{E}[d(s, s')] - \mathbb{E}[d(y, y')],$$

where $d(\cdot, \cdot)$ is DSC or intersection over union, s, s' are samples generated by the model, and y, y' are real segmentations sampled from the data set.

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- Correlation between amount of uncertainty and the error on test samples.
- Expected Calibration Error (ECE)

$$ECE(\mathcal{D}; \theta) \triangleq \mathbb{E}_r[|\mathbb{E}_c[r][c] - r|],$$

where r is the confidence the model assigns to samples and c is an indicator function that says whether the prediction is correct.

Outline

Quantifying uncertainty in model predictions

Predictions of machine learning models, like any other models, may be uncertain. Quantifying this uncertainty for medical imaging applications can be crucial. How do we do this for deep learning models?

- 1 The problem
- 2 Mathematical treatment
 - Posterior distribution
 - Approximations
 - Evaluation
- 3 Examples
- 4 Unintuitive behavior in high dimensions

Uncertainty over θ and y given x

[Gal and Ghahramani 2016, Tanno et al. 2020, Tanno et al. 2017]

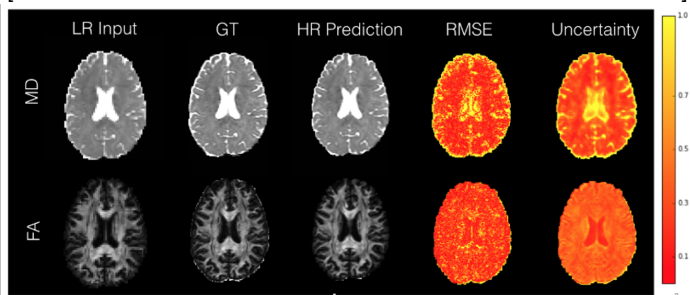


Figure 8: Comparison between voxel-wise RMSE and predictive uncertainty maps for FA and MD computed on a HCP test subject (min-max normalised for MD and FA separately). Low-res input, ground truth and the mean of high-resolution predictions are also shown.

They combine (1) *variational drop-out* to sample θ given $\mathcal{D}, \mathcal{M}$ to approximate $p(\theta|\mathcal{D}, \mathcal{M})$ and (2) a pixel-wise distribution to model $p(y|x, \theta, \mathcal{M})$.

Segmentation

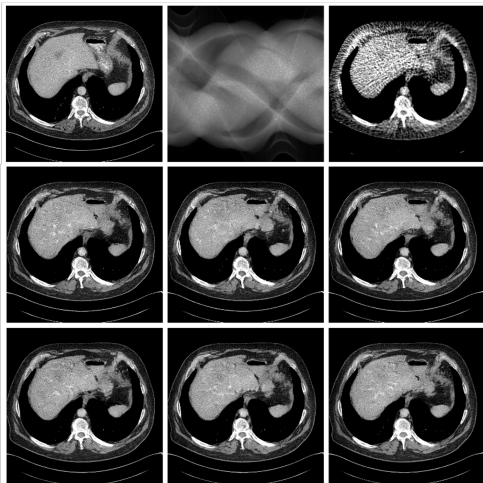
[Kohl et al., 2018, Baumgartner et al. 2019]



They define a probabilistic latent space model to approximate $p(y|x, \theta, \mathcal{M})$, and train with data set that contains multiple annotations for a given image.

Sampling

[Adler and Öktem 2018]



They define a sampler to model $p(y|x, \theta, \mathcal{M})$ that uses an adversarial loss to make sure outputs are realistic and also satisfy the data.

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High dimensional distributions are counter intuitive¹

$$p(y|x) = \int_{\mathcal{M}} \int_{\mathcal{D}} \int_{\theta} p(y|x, \theta, \mathcal{M}) dp(\theta|\mathcal{D}, \mathcal{M}) dp(\mathcal{D}) dp(\mathcal{M})$$

¹Joint work with Tareen Dawood, Roberto Cagnotti, Kilian Zepf and Aasa Feragen.
[Cagnotti 2023] and [Zepf et al. 2026]

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(a) Original



(b) Noisy



(c) MAP estimate

Deep decoder denoising with x being noisy and $\hat{y}$ being noise-free image estimate.

$$\hat{y} = f(z|\theta^*), \quad \theta^* = \arg_{\theta} \max \|x - f(z|\theta)\|_2^2, \quad z \sim \mathcal{N}(0, \Sigma)$$

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Sampling model parameters - “epistemic uncertainty”

$$\hat{y} = f(z|\theta^*), \quad \theta^* = \arg_{\theta} \max \|x - f(z|\theta)\|_2^2, \quad z \sim \mathcal{N}(0, \Sigma)$$

Instead, we could formulate the probabilistic model

$$p(\theta|x, z, \mathcal{M}) \propto p(x|\theta, z, \mathcal{M})p(\theta|\mathcal{M}), \quad p(x|\theta, z, \mathcal{M}) = \mathcal{N}(f(z|\theta), \mathbf{I})$$

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and sample [Sidak 1967]

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(a) Original



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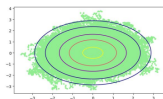
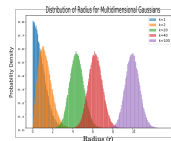


(c) MAP estimate

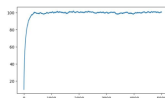


(d) Posterior sample

Typical sets



(a) Visualizations of the samples and theoretical contour lines. Here only the first 2 dimensions of each sample are plotted.

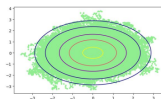
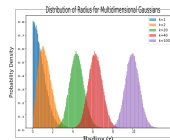


(b) L_2 norm of the first 5'000 samples

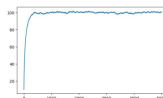
For an isotropic unit Gaussian distribution in d dimensions, the expected distance of any sample from the mean is [Blum et al. 2024]

$$\mathbb{E}[|x - \mu|] = \sqrt{d}.$$

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For an isotropic unit Gaussian distribution in d dimensions, the expected distance of any sample from the mean is [Blum et al. 2024]

$$\mathbb{E}[|x - \mu|] = \sqrt{d}.$$

As the dimension increases, the samples start to move away from the mode of the distribution.

Samples moving away from high likelihood areas



(a) Original



(b) Noisy



(c) MAP estimate



(d) Posterior sample

In this instance

$$p(\theta|x, z, \mathcal{M}) \propto p(x|\theta, z, \mathcal{M})p(\theta|\mathcal{M}), \quad p(x|\theta, z, \mathcal{M}) = \mathcal{N}(f(z|\theta), \mathbf{I})$$

More generally

$$p(\theta|\mathcal{D}, \mathcal{M}) \propto p(\mathcal{D}|\theta, \mathcal{M})p(\theta|\mathcal{M})$$

where $p(\mathcal{D}|\theta, \mathcal{M})$ is the likelihood term, in other words **the task-specific loss you are minimizing!** Bounded sampling [Zepf et al. 2026] may help but does not provide a great solution.

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